ARM Models

	Fixed effects only	Random and fixed effects
Linear regression	398	168
Logistic regression	192	10

Models

	Cross-covariance estimate	Variance estimate
Log scale parameter	191	62
Regression parameter	325	190

Covariances to estimate

Table 1: Models from [?]

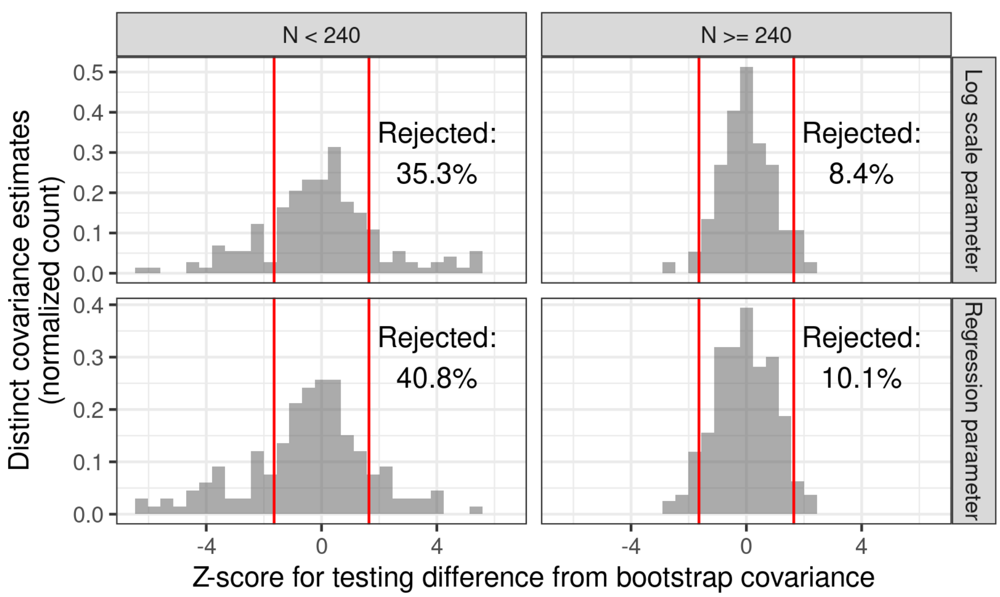

Figure 1: The distribution of the z-statistics Z. Red lines indicate the boundaries of a two-sided normal test for significance with level 0.1, and "Rejected" counts the number of covariances in both sides of the rejection region. The "Log scale parameter" panel includes all variances or covariances that involve at least one log scale parameter.

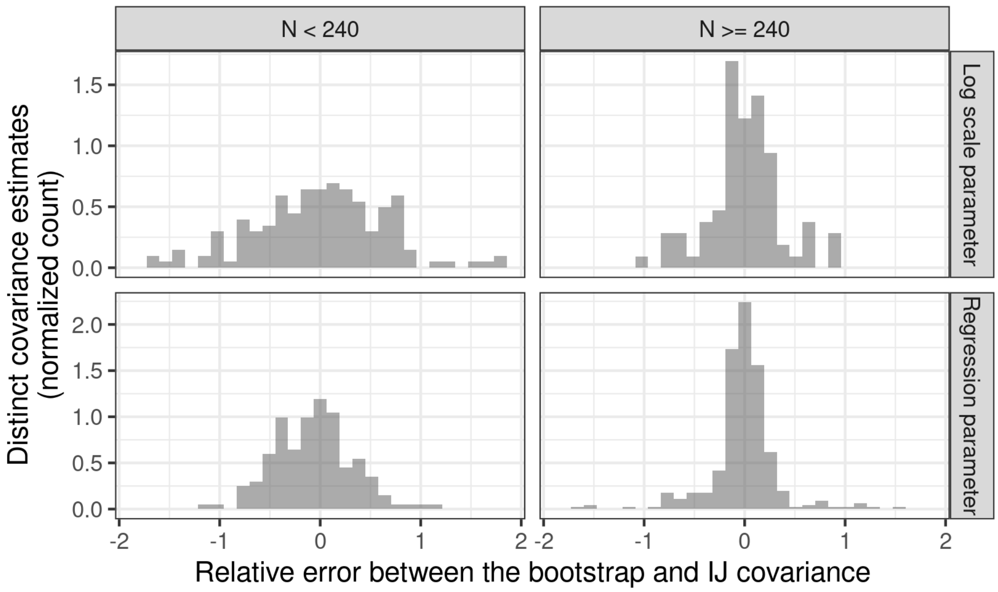

Figure 2: The distribution of the relative errors Δ^{IJ}. The "Log scale parameter" panel includes all variances or covariances that involve at least one log scale parameter.

Singular Simulation

MRP

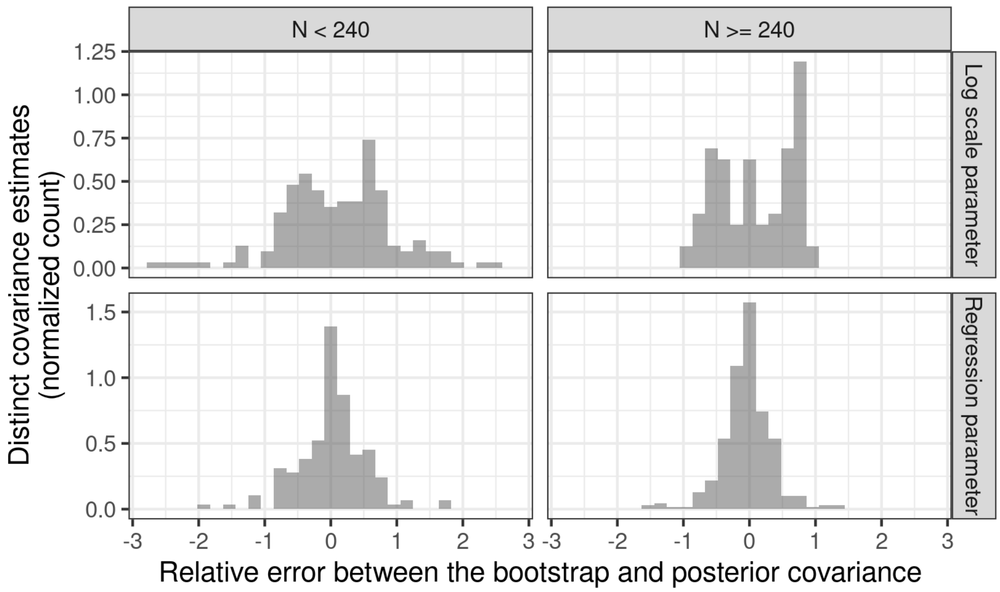

Figure 3: The distribution of the relative errors Δ^{Bayes}. The "Log scale parameter" panel includes all variances or covariances that involve at least one log scale parameter.

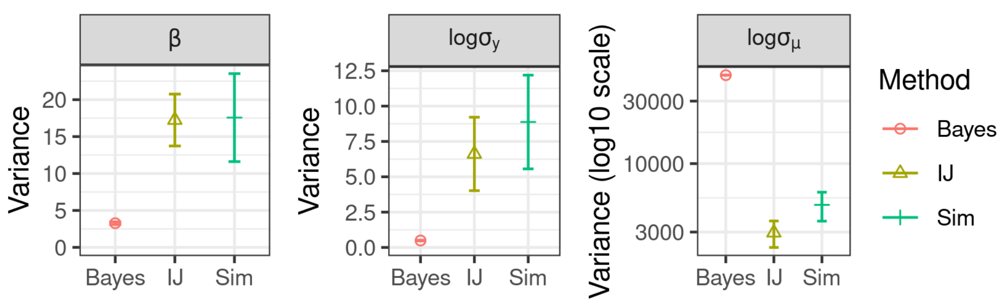

Figure 4: The difference from the simulation, IJ, and Bayesian covariances. Standard errors include variation due to both Monte Carlo and, in the case of the IJ and simulation, frequentist error.

	Chapter	Model	Dataset	Number of parameters
1	ARM_Ch.10	ideo_reparam	ideo	15
2	ARM_Ch.10	sesame_multi_preds_3a	sesame	45
3	ARM_Ch.10	sesame_multi_preds_3b	sesame	45
4	ARM_Ch.10	sesame_one_pred_2b	sesame	6
5	ARM_Ch.10	sesame_one_pred_a	sesame	6
6	ARM_Ch.10	sesame_one_pred_b	sesame	6
7	ARM_Ch.12	radon_complete_pool	radon	6
8	ARM_Ch.12	radon_group	radon	15
9	ARM_Ch.12	radon_group_chr	radon	15
10	ARM_Ch.12	radon_intercept	radon	6
11	ARM_Ch.12	radon_intercept_chr	radon	6
12	ARM_Ch.12	radon_no_pool	radon	10
13	ARM_Ch.12	radon_no_pool_chr	radon	10
14	ARM_Ch.13	pilots	pilots	10
15	ARM_Ch.13	radon_inter_vary	radon	21
16	ARM_Ch.13	radon_vary_si	radon	10
17	ARM_Ch.14	election88	election	10
18	ARM_Ch.20	hiv	hiv	10
19	ARM_Ch.20	hiv_inter	hiv	15
20	ARM_Ch.23	electric	electric	10
21	ARM_Ch.23	electric_1a	electric	15
22	ARM_Ch.23	electric_1b	electric	15
23	ARM_Ch.23	electric_multi_preds	electric	10
24	ARM_Ch.23	electric_one_pred	electric	6
25	ARM_Ch.3	kidiq_interaction	kidscore	15
26	ARM_Ch.3	kidiq_multi_preds	kidscore	10
27	ARM_Ch.3	kidscore_momhs	kidscore	6
28	ARM_Ch.3	kidscore_momiq	kidscore	6
29	ARM_Ch.4	logearn_height	earnings	6
30	ARM_Ch.4	logearn_height_male	earnings	10
31	ARM_Ch.4	logearn_interaction	earnings	15
32	ARM_Ch.4	logearn_logheight	earnings	10
33	ARM_Ch.4	mesquite_log	mesquite	36
34	ARM_Ch.4	mesquite_va	mesquite	15
35	ARM_Ch.4	mesquite_vas	mesquite	36
36	ARM_Ch.4	mesquite_vash	mesquite	28
37	ARM_Ch.4	mesquite_volume	mesquite	15
38	ARM_Ch.5	nes2000_vote	vote	3
39	ARM_Ch.5	separation	separation	3
40	ARM_Ch.5	wells_d100ars	wells	6
41	ARM_Ch.5	wells_daae_c	wells	21
42	ARM_Ch.5	wells_dae	wells	10
43	ARM_Ch.5	wells_dae_c	wells	15
44	ARM_Ch.5	wells_dae_inter	wells	15
45	ARM_Ch.5	wells_dae_inter_c	wells	28

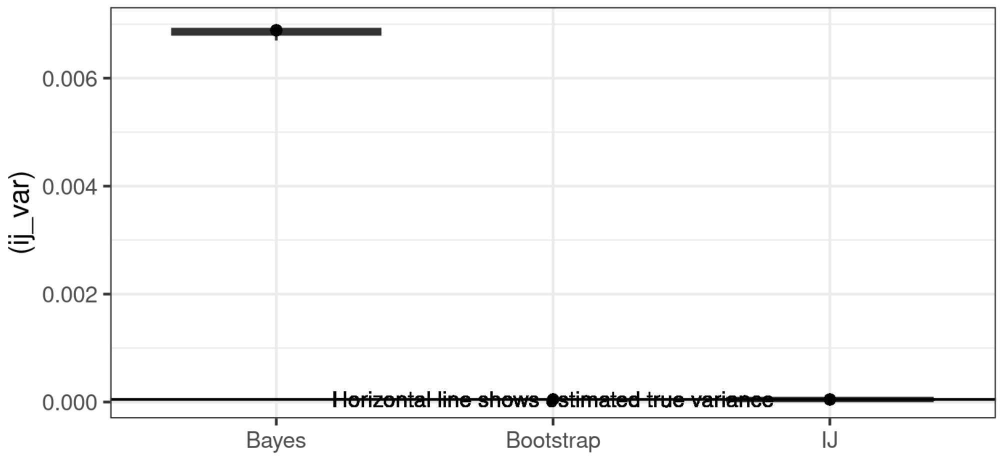

Figure 5: Covariance estimates for the MrP model. The dots are the estimates computed on the single dataset chosen as the basis for bootstrapping. The boxplots show the spread of the IJ and posterior covariance across simulations, which includes both frequentist and MCMC variability. The horizontal line shows the true sampling variance as estimated by simulation.

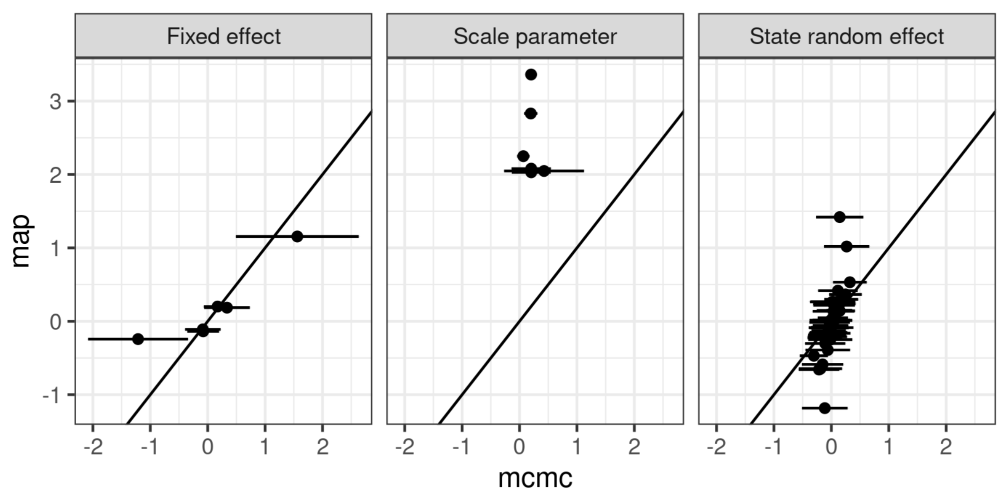

Figure 6: MAP versus MCMC for the MrP model

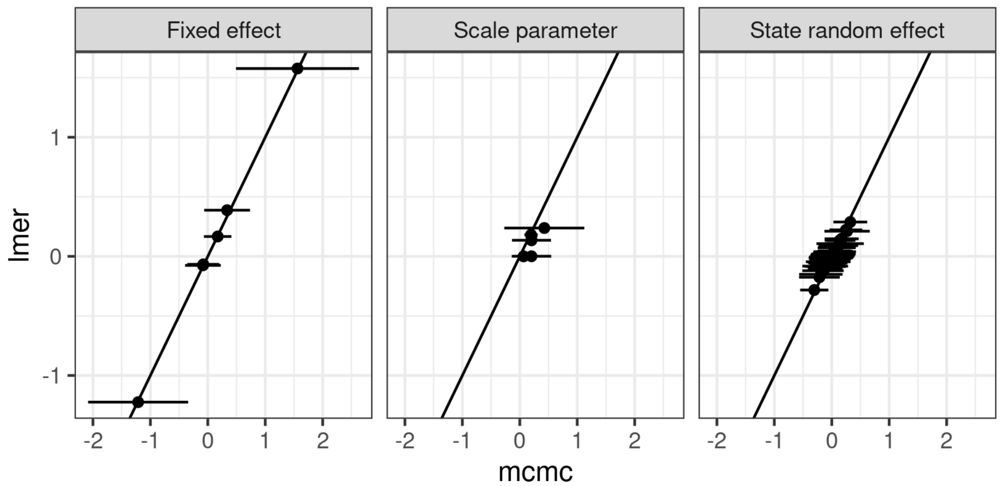

Figure 7: LMER versus MCMC for the MrP model

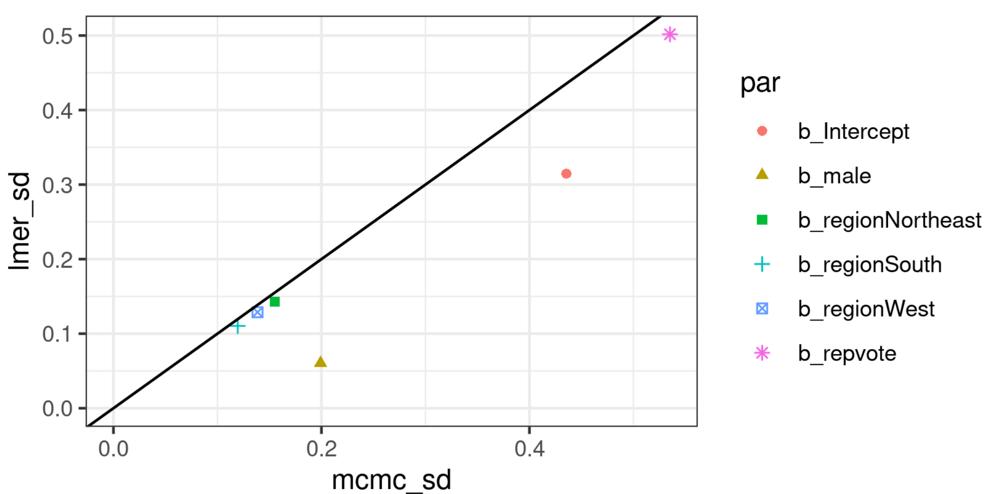

Figure 8: LMER versus MCMC standard errors for the MrP model